

حلول أسئلة السنوات

٢٠١٨ / ٢٠١٧

لمادة

Calculus ١

للأستاذ : أحمد عرفه

٠٧٨٦٥٤٣٦٦٥

كل سؤال يحوي اقتران أسّي e^x أو لوغاريتمي $\ln x / \log x$ احتفظ به لكالكولس ٢ .

Calculus 1

Midterm "2017/2018-1st semester"

Q1) The range of the function $f(x) = \frac{1}{\sqrt{4-x^2}}$ is

- a) $(0, 2)$ b) $[0, 2]$ c) $(\frac{1}{2}, \infty)$ d) $[\frac{1}{2}, \infty)$

Q2) Given that $f(x) = \frac{1}{x-3}$ and $g(x) = \frac{1}{x}$ then the domain of the function $f \circ g$ is

- a) $\mathbb{R} \setminus \{0, \frac{1}{3}\}$ b) $\mathbb{R} \setminus \{0\}$ c) $\mathbb{R} \setminus \{\frac{1}{3}\}$ d) $\mathbb{R} \setminus \{0, 3\}$

Q3) (The greatest integer less than or equals x) The range of $f(x) = 2[x]$ is

- a) $\mathbb{R} \setminus \{0, \pm 1, \pm 2, \pm 3, \dots\}$ b) $\{0, \pm 1, \pm 2, \pm 3, \dots\}$ c) $(-\infty, \infty)$ d) $\{0, \pm 2, \pm 4, \pm 6, \dots\}$

Q4) If the domain of the function $y = f(x)$ is $[2, 3)$ then the domain of

$g(x) = f(3 - x)$ is

- a) $[2, 3)$ b) $(0, 1]$ c) $[0, 1)$ d) $(2, 3]$

Q5) Given that $f(x) = x^2$ and $g(x) = \begin{cases} 2x & x < 4 \\ x+1 & x \geq 4 \end{cases}$ then $(f \circ g)(x) =$

- a) $\begin{cases} 4x^2 & x < 4 \\ (x+1)^2 & x \geq 4 \end{cases}$ c) $\begin{cases} 4x^2 & x < 16 \\ (x+1)^2 & x \geq 16 \end{cases}$
b) $\begin{cases} 4x^2 & x \leq 4 \\ (x+1)^2 & x > 4 \end{cases}$ d) $\begin{cases} 4x^2 & x \leq 16 \\ (x+1)^2 & x > 16 \end{cases}$

Q6) The domain of the function $f(x) = \ln(5 - |7x + 3|)$ is

- a) $[-\frac{8}{7}, \frac{2}{7}]$ b) $\mathbb{R} \setminus [-\frac{8}{7}, \frac{2}{7}]$ c) $(-\frac{8}{7}, \frac{2}{7})$ d) $\mathbb{R} \setminus (-\frac{8}{7}, \frac{2}{7})$

Q7) The x -intercept of the function $f(x) = -2 + \ln(3x + 1)$ is

- a) 0 b) $\frac{1}{3}e^2$ c) $\frac{1}{3}(e^2 - 1)$ d) $\frac{1}{3}(1 - e^2)$

Q8) The solution set of the inequality $e^{2x} > 5e^x$ is

- a) $(-\infty, \ln \sqrt{5})$ b) $(\ln \sqrt{5}, \infty)$ c) $(0, \frac{1}{2} \ln 5)$ d) $(-\frac{1}{2} \ln 5, 0)$

Q9) The domain of $f(x) = \cos^{-1}(3x + 1)$ is

- a) $[-1, 1]$ b) $[-\frac{1}{2}, 0]$ c) $[-\frac{2}{3}, 0]$ d) $[0, \frac{2}{3}]$

Q10) The range of the function $f(x) = 4 \sin^{-1}(x^2)$ is

- a) $[-\frac{\pi}{2}, \frac{\pi}{2}]$ b) $[-2\pi, 2\pi]$ c) $[0, 2\pi]$ d) $[-2\pi, 0]$

Q11) Given that $f(x) = \csc^{-1} x$ then $f(2) =$

- a) $\frac{1}{\sin 2}$ b) $\frac{1}{\cos 2}$ c) $\cos^{-1}(\frac{1}{2})$ d) $\sin^{-1}(\frac{1}{2})$

Q12) If the $f(x) = x^2 - 4x + 7$, $x \in [k, \infty)$ is one to one function, then the smallest possible value of k is

- a) 4 b) 2 c) -2 d) -4

Q13) Given that $f(x) = \pi + \cot^{-1}(x - 2)$ then $f(2) =$

- a) $\frac{\pi}{2}$ b) $\frac{3}{2}\pi$ c) π d) *undefined*

Q14) The value of $4 \sin^{-1}(\frac{\sqrt{5}}{19}) + 4 \cos^{-1}(\frac{\sqrt{5}}{19}) =$

- a) π b) $\frac{1}{2}\pi$ c) $\frac{1}{4}\pi$ d) 2π

Q15) $\lim_{x \rightarrow -\infty} \left(\frac{\sqrt{49x^2 + x + 1} + 2x + 9}{6x + 4} \right) =$

- a) $-\frac{5}{6}$ b) $\frac{3}{2}$ c) ∞ d) $-\infty$

Q16) $\lim_{x \rightarrow 2^-} \left(\frac{1}{x-2} - \frac{12}{x^3-8} \right) =$
 a) 1 b) -1 c) $\frac{1}{2}$ d) $\frac{1}{2}$

Q17) $\lim_{x \rightarrow 4^+} \left(\frac{x^2 - 6x + 8}{\sqrt{x^2 - 8x + 16}} \right) =$
 a) 2 b) -2 c) 0 d) does not exist

Q18) If $y = 0$ and $x = 2$ are horizontal and vertical asymptotes (respectively) of the function $f(x) = \frac{ax^3 + x(3x+1)}{x^3 - b}$ then $a + b =$
 a) 11 b) 8 c) -1 d) 5

Q19) The function $f(x) = x^2 \ln |x|$ is
 a) even b) odd c) neither odd nor even d) positive for all $x \in \text{Domain}(f)$

Q20) $f(x) = \begin{cases} 2-x & x \leq 0 \\ 1+e^{-x} & x \geq 0 \end{cases}$ then $f^{-1}(x) =$
 a) $\begin{cases} x-2 & x > 2 \\ -\ln(x-1) & 1 < x \leq 2 \end{cases}$ c) $\begin{cases} 2-x & x > 2 \\ -1+\ln x & 1 < x \leq 2 \end{cases}$
 b) $\begin{cases} 2-x & x > 2 \\ -\ln(x-1) & 1 < x \leq 2 \end{cases}$ d) $\begin{cases} x-2 & x > 2 \\ -1+\ln x & 1 < x \leq 2 \end{cases}$

Ahmad Arafah

0786543665

Good Luck

Q1:~

$$\begin{aligned} \text{Sol: } x^2 &> 0 \\ -x^2 &\leq 0 \\ 4 - x^2 &\leq 4 \\ \sqrt{4 - x^2} &\leq 2 \end{aligned}$$

$$\frac{1}{\sqrt{4 - x^2}} > \frac{1}{2}$$

$$\left[\frac{1}{2}, \infty\right) \quad (d)$$

$$Q_2: f(x) = \frac{1}{x-3}, g(x) = \frac{1}{x}$$

$$D_{f \circ g} = ?$$

$$\text{Sol: } (f \circ g)(x) = f(g(x))$$

$$= f\left(\frac{1}{x}\right) = \frac{1}{\frac{1}{x} - 3}$$

$$\frac{1}{x} - 3 = 0 \Rightarrow \frac{1}{x} = 3 \Rightarrow x = \frac{1}{3}$$

$$D_{f \circ g} = \mathbb{R} - \left\{\frac{1}{3}\right\}$$

$$D_{f \circ g} = D_{f(g(x))} \cap D(g(x))$$

$$= (\mathbb{R} - \left\{\frac{1}{3}\right\}) \cap (\mathbb{R} - \{0\})$$

$$= \mathbb{R} - \left\{0, \frac{1}{3}\right\} \quad (a)$$

$$Q_3: \text{Range } [x] = \{0, \pm 1, \pm 2, \pm 3, \dots\}$$

$$\text{Range } 2[x] = \{0, \pm 2, \pm 4, \pm 6, \dots\}$$

(d)

$$Q_4: Df(x) = [2, 3) \text{ then}$$

$$Df(3-x) \text{ is ?}$$

$$\text{Sol: } 2 \leq 3-x < 3$$

(-3)

(-3)

(-3)

$$\Rightarrow -1 \leq -x < 0$$

$$1 \geq x > 0 \Rightarrow (0, 1] \quad (b)$$

Q5:~

$$g(x) = 2x \rightarrow f(2x) = (2x)^2 = 4x^2$$

$$g(x) = x+1 \rightarrow f(x+1) = (x+1)^2$$

$$(f \circ g)(x) = f(g(x))$$

$$= (2x)^2, \quad x < 4$$

$$= (x+1)^2, \quad x \geq 4$$

$$= \begin{cases} 4x^2 & x < 4 \\ (x+1)^2 & x \geq 4 \end{cases}$$

(a)

Q6:~

زوج

Q7:

The x -intercept $\rightarrow y=0$

$$0 = -2 + \ln(3x+1)$$

$$2 = \ln(3x+1)$$

$$e^2 = 3x+1$$

$$e^2 - 1 = 3x$$

$$x = \frac{e^2 - 1}{3} \quad (C)$$

Q8: حروف

$$Q9: -1 \leq 3x+1 \leq 1$$

$$-2 \leq 3x \leq 0$$

$$-\frac{2}{3} \leq x \leq 0$$

$$\left[-\frac{2}{3}, 0\right]$$

$$Q10: \text{Range } \sin^{-1}(x^2) = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\text{Range } 4\sin^{-1}(x^2) = 4\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$= [-2\pi, 2\pi]$$

(B)

$$Q11: f(x) = \csc^{-1} x, \text{ then } f(2) =$$

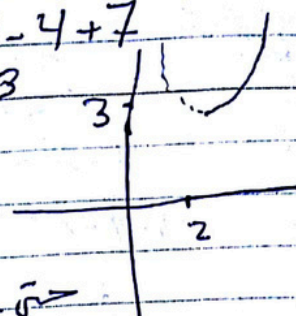
$$\text{Sol: } \csc^{-1} 2 = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

(D)

$$Q12: f(x) = x^2 - 4x + 7, \\ x \in [k, \infty) \text{ is } 1-1$$

$$\text{Sol: } x^2 - 4x + 4 - 4 + 7 \\ = (x-2)^2 + 3$$

$$(k=2)$$



حتى يكون 1-1 يجب
أن يكون مجاله
[2, ∞)

$$Q13: f(x) = \pi + \cot^{-1}(x-2) \\ \text{then } f(2) =$$

$$f(2) = \pi + \cot^{-1}(0) \\ = \pi + \frac{\pi}{2} = \frac{3\pi}{2} \quad (B)$$

$$Q14: \text{Rule } \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

$$\text{Sol: } 4\sin^{-1}\left(\frac{\sqrt{5}}{19}\right) + 4\cos^{-1}\left(\frac{\sqrt{5}}{19}\right) \\ = 4\left(\sin^{-1}\left(\frac{\sqrt{5}}{19}\right) + \cos^{-1}\left(\frac{\sqrt{5}}{19}\right)\right) \\ = 4\left(\frac{\pi}{2}\right) = 2\pi \quad (d)$$

$$Q15: \lim_{x \rightarrow \infty} \frac{\sqrt{49x^2 + x + 1} + 2x + 9}{6x + 4}$$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{49|x| + 2x}}{6x}$$

نأخذ أكبر قوة
بالسطر مع أكبر قوة بالقامع مع مراعاة
أن بالسطر حركنا لها نفس القوة
 $\sqrt{49x^2 + x + 1} + 2x + 9$

$$\sqrt{x^2} = |x|$$

$$\begin{array}{l} x \rightarrow \infty \quad \text{إن } ١٦ \\ |x| \rightarrow -x \quad \text{إذن} \end{array}$$

$$\lim_{x \rightarrow \infty} \frac{-7x + 2x}{6x} = \lim_{x \rightarrow \infty} \frac{-5x}{6x} = -\frac{5}{6}$$

$$Q16: \lim_{x \rightarrow 2^-} \left(\frac{1}{x-2} - \frac{12}{x^3-8} \right)$$

$$\text{Sol } x^3-8 = (x-2)(x^2+2x+4)$$

$$\lim_{x \rightarrow 2^-} \frac{x^2+2x+4-12}{x^3-8}$$

$$= \lim_{x \rightarrow 2^-} \frac{x^2+2x-8}{x^3-8}$$

$$= \lim_{x \rightarrow 2^-} \frac{(x-2)(x+4)}{(x-2)(x^2+2x+4)}$$

$$= \frac{6}{12} = \frac{1}{2}$$

$$Q17: \lim_{x \rightarrow 4^+} \frac{x^2-6x+8}{\sqrt{(x-4)^2}}$$

$$\text{Sol } \lim_{x \rightarrow 4^+} \frac{(x-4)(x-2)}{|x-4|} = \frac{+}{+}$$

$$= \lim_{x \rightarrow 4^+} \frac{(x-4)(x-2)}{x-4} = 4-2 = 2 \quad (a)$$

$$Q18: y=0 \text{ and } x=2$$

$$\text{Sol } y=0$$

$$\lim_{x \rightarrow \infty} \frac{ax^3+3x^2+x}{x^3-b} = 0$$

$$\lim_{x \rightarrow \pm \infty} \frac{a}{1} = 0$$

$$a=0$$

$$x=2$$

$$\Rightarrow \lim_{x \rightarrow 2} \frac{x(3x+1)}{x^3-b} = \pm \infty$$

$$\frac{2(7)}{8-b} = \frac{14}{8-b} = \pm \infty$$

$$\text{if } b=8$$

$$\therefore a+b=0+8=8 \quad (b)$$

$$Q19: f(-x) = (-x)^2 \ln|x|$$

$$= x^2 \ln x$$

$$= f(x)$$

$$\therefore f \text{ is even } (a)$$

Q20

$$y = 2 - x$$

$$x = 2 - y$$

$$\Rightarrow y = 2 - x$$

$$f^{-1}(x) = 2 - x$$

$$y = 1 + e^{-x}$$

$$x = 1 + e^{-y}$$

$$x - 1 = e^{-y}$$

$$\ln(x-1) = \ln(e^{-y})$$

$$\ln(x-1) = -y$$

$$y = -\ln(x-1)$$

$$f^{-1}(x) = -\ln(x-1)$$

$$f^{-1}(x) = \begin{cases} 2 - x & , x > 2 \\ -\ln(x-1) & , 1 < x \leq 2 \end{cases}$$

Balqa Al-salt 2015

Q1) Fill The Following Blanks :

- 1- If $f(3x) = 9x + 5$, then $f(x) =$
- 2- The graph $f(x) = x \sin x$ is symmetric around
- 3- Is the function $f(x) = \frac{x^5|x|}{x^4-3}$ even, odd or neither
- 4- The graph of $\sqrt{x+2} - 3$ is obtained from the graph of $\sqrt{x}$ by _____ and _____
- 5- The graph of $-x^2 + 8x - 16$ is obtained from the graph of x^2 by _____ and _____
- 6- If $f(x) = \sqrt{x-1}$, $g(x) = \frac{1}{x^2-5}$, then the $D_{g \circ f}$ is
- 7- If $f(x) = 2x^3 - 3x^2 + 1$, and $f^{-1}(x) = 1$, then $x =$
- 8- $\log_x(2x+3) = 2$, then $x =$
- 9- The domain of $f(x) = 1 + \sqrt{16 - x^2}$ is
- 10- If the domain and range of $f(x)$ are $[-1, 3]$ and $(0, 4]$, respectively, then the domain of $g(x) = 1 - 3f(3 - 2x)$ is
- 11- The range of $f(x) = \sin^{-1}(3x+1) + \frac{\pi}{2}$ is
- 12- $\sin^{-1}(\sin \frac{9\pi}{7})$

Q2) If $f(x) = \frac{e^x}{1-5e^x}$. Find

- a) Domain (f)
- b) f^{-1}
- c) Range f

Good Luck

Ahmad Arafah

2015

Q1)

$$1 - y = 3x$$

$$\rightarrow x = \frac{y}{3}$$

$$f(y) = 9\left(\frac{y}{3}\right) + 5$$

$$f(y) = 3y + 5$$

$$\therefore f(x) = 3x + 5$$

3- $f(x)$ is odd

$$f(-x) = \frac{(-x)^5 |1-x|}{(-x)^4 - 3} = \frac{-x^5 |x|}{x^4 - 3}$$

$$-f(x) = \frac{-x^5 |x|}{x^4 - 3}$$

$$\therefore f(-x) = -f(x)$$

$$5- -(x^2 - 8x - 16)$$

$$-(x^2 - 8x + 16 - 16 - 16)$$

$$-((x-4)^2 - 32)$$

$$32 - (x-4)^2$$

4 Units to the right and 32 Units upward and reflection about x-axis.

7-

$$x = f(1)$$

$$= 2(1) - 3(1) + 1$$

$$x = 0$$

2- around y-axis

$$f(-x) = -x \sin(-x)$$

Since $\sin x$ is odd.

$$= -x \cdot -\sin(x)$$

$$= x \cdot \sin x$$

رفع مكان

$$x \rightarrow -x$$

4- Two Units to the left and three Units to the down.

$$\sqrt{x+2} - 3$$

دلتا 2 ←
← اد ←

← اد ←
← اد ←

6- Dgof

$$g \circ f(x) = g(f(x)) = g(\sqrt{x-1})$$

$$= \frac{1}{x-1-5} = \frac{1}{x-6}$$

$$D_{g \circ f} = D_{g \circ f} \cap D_f$$

$$= (\mathbb{R} - \{6\}) \cap [1, \infty)$$

$$= [1, \infty) - \{6\}$$

8-

$$x^2 = 2x + 3$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1)$$

$$x=3, x=-1$$

9-

$$16 - x^2 \geq 0$$

$$x^2 \leq 16$$

$$\sqrt{x^2} \leq \sqrt{16}$$

$$|x| \leq 4$$

$$-4 \leq x \leq 4$$

$$Df(x) = [-4, 4]$$

10-

$$-1 \leq 3 - 2x \leq 3$$

$$\begin{matrix} (-3) & (-3) & (-3) \end{matrix}$$

$$-4 \leq -2x \leq 0$$

عند القسمة على سالبة نغيث المتباينة

$$\frac{-4}{-2} \geq \frac{-2x}{-2} \geq \frac{0}{-2}$$

$$0 \leq x \leq 2$$

$$Dg(x) = [0, 2]$$

$$11- R \sin^{-1} x = \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

$$R \sin^{-1}(3x+1) + \frac{\pi}{2} = \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] + \frac{\pi}{2}$$

هذا ما يؤثر

$$= [0, \pi]$$

12-

$$\sin^{-1} \left(\sin \frac{6\pi}{7} \right)$$

$$= \sin^{-1} \left(\sin \frac{\pi}{7} \right) = \frac{\pi}{7}$$

$$\text{Since } \frac{\pi}{7} \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

$$Q2) a) Df =$$

المقام

$$1 - 5e^x = 0$$

$$5e^x = 1$$

$$e^x = \frac{1}{5}$$

$$x = \ln\left(\frac{1}{5}\right)$$

$$Df = (R \cap R) - \left\{ \ln\left(\frac{1}{5}\right) \right\}$$

$$= R - \left\{ \ln\left(\frac{1}{5}\right) \right\}$$

$$b) y = \frac{e^x}{1-5e^x}$$

$$\text{Sol. } x = \frac{e^y}{1-5e^y}$$

$$x - 5x e^y = e^y$$

$$x = e^y + 5x e^y$$

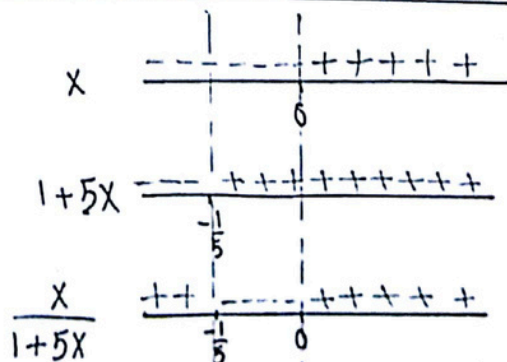
$$x = e^y (1 + 5x)$$

$$e^y = \frac{x}{1+5x} \Rightarrow y = \ln\left(\frac{x}{1+5x}\right)$$

$$\therefore f^{-1}(x) = \ln\left(\frac{x}{1+5x}\right)$$

$$c) Rf = Df^{-1}$$

$$= \left(-\infty, -\frac{1}{5}\right) \cup (0, \infty)$$

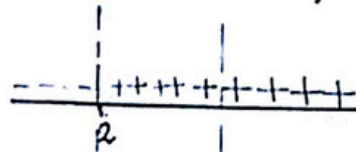


Q1)

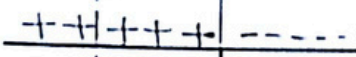
1-

$$D f(x) = [2, 5]$$

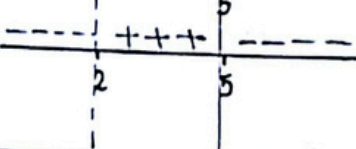
$$2x - 4$$



$$5 - x$$



$$\frac{2x-4}{5-x}$$



2-

$$-1 \leq \cos x \leq 1$$

$$-1 \leq -\cos x \leq 1$$

$$4 - 1 \leq 4 - \cos x \leq 4 + 1$$

$$3 \leq 4 - \cos x \leq 5$$

$$\frac{1}{5} \leq \frac{1}{4 - \cos x} \leq \frac{1}{3}$$

$$\frac{3}{5} \leq \frac{3}{4 - \cos x} \leq 1$$

$$R f(x) = \left[\frac{3}{5}, 1\right]$$

3-

$$f(-x) = \frac{\sin(-x)}{(-x)^3 - (-x)^2} = \frac{-\sin x}{-x^3 - x^2} \neq f(x)$$

$$-f(x) = \frac{-\sin(x)}{x^3 - x^2} \neq f(-x)$$

∴ neither.

4- about y-axis

$$y^3 = 2(-x)^4 - 5(-x)^6$$

$$y^3 = 2x^4 - 5x^6$$

5-

$$f(g(x)) = (|x| + 7)^{3/5}$$

$$g(x) = |x| + 7$$

$$f(x) = x^3$$

6-

$$2 \leq 3x + 1 \leq 5$$

$$-1 \quad -1 \quad -1$$

$$1 \leq 3x \leq 4$$

$$\frac{1}{3} \leq x \leq \frac{4}{3}$$

$$Dg(x) = \left[\frac{1}{3}, \frac{4}{3}\right]$$

7. like هذا

$$x^2 - 2x + 2 \Rightarrow x^2 - 4x + 3$$

نقل الـ 2

$$(x-1)^2 + 1 \Rightarrow (x-2)^2 - 1$$

كيف ينظر

One unit to the left and two units upward.

8- $y = \frac{3x-4}{2-x}$

$$x = \frac{3y-4}{2-y}$$

$$2x - xy = 3y - 4$$

$$2x + 4 = 3y + xy$$

$$2x + 4 = y(3 + x)$$

$$y = \frac{2x+4}{3+x}$$

$$\therefore f^{-1}(x) = \frac{2x+4}{3+x}$$

9- y -axis

10- $g \circ f(1) = g(f(1))$

$$= g(-3)$$

g is even

f odd

$$= g(3)$$

$$= 6$$

$$f(-1) = 3$$

$$-f(1) = 3$$

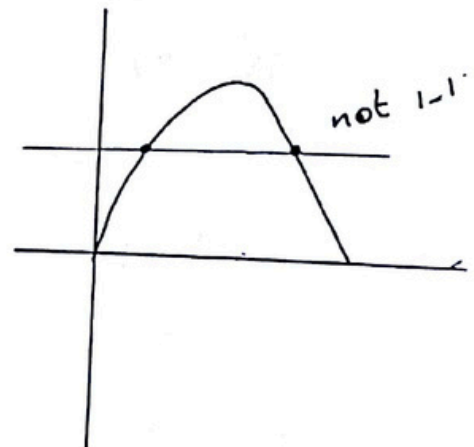
$$f(1) = -3$$

11- $f \circ g(x) = f(g(x))$

$$= f(\sqrt{3x-6}) = \frac{1}{2-\sqrt{3x-6}}$$

Q2) $\square$ $\sin x$, $0 < x < \pi$

لأن مجال $\sin x$ هو $[-\frac{\pi}{2}, \frac{\pi}{2}]$
حتى يكون لها انعكاس



Mid - Exam 21 - 3 - 2017

Q1) Evaluate the limit

$$\lim_{x \rightarrow 2} \frac{1}{4x^2 - 16}$$

Q2) Find the value(s) of x (if any) that satisfy

$$\sin(3x) = 0, \quad x \in [0, \frac{\pi}{3}]$$

Q3) Find the value(s) of x (if any) that satisfy

$$\ln(x - 1) = 0$$

Q4) Evaluate the limit

$$\lim_{x \rightarrow 0} \frac{\sin(3x^6)}{\tan^2(x^3)}$$

Q5) Evaluate the limit

$$\lim_{x \rightarrow 3} \frac{\sqrt{x+1} - 2}{x - 3}$$

Q6) Evaluate the limit

$$\lim_{x \rightarrow \infty} \tan^{-1}\left(\frac{1}{x^2 + 1}\right)$$

Q7) Study the continuity of the function $f(x) = \ln(x^2 + e^x)$

Q8) Study the continuity of the following function at $x = 0$

$$f(x) = \begin{cases} e^{2x} - 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Good Luck

Ahmad Arafah

0786543665

$$Q_1) \lim_{x \rightarrow 2^+} \frac{1}{4x^2 - 16} = \frac{1}{0^+} = \infty$$

$$\lim_{x \rightarrow 2^-} \frac{1}{4x^2 - 16} = \frac{1}{0^-} = -\infty$$

$$\therefore \lim_{x \rightarrow 2} \frac{1}{4x^2 - 16} \text{ d.n.e}$$

$$Q_2) \sin(3x) = 0$$

$$3x = 0, \pi, 2\pi$$

$$x = 0, \frac{\pi}{3}, \frac{2\pi}{3}$$

$$x = \{0, \frac{\pi}{3}\}$$

$$Q_3) \ln(x-1) = 0$$

$$e^{\ln(x-1)} = e^0 \Rightarrow x-1 = 1$$

$$x = 1 + 1 = 2$$

$$x = 2$$

$$Q_4) \lim_{x \rightarrow 0} \frac{\sin(3x^6)}{\tan^2(x^3)}$$

Sol: x^6 is small.

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\frac{\sin(3x^6)}{x^6}}{\frac{\tan^2(x^3)}{x^6}} &= \frac{\lim_{x \rightarrow 0} \frac{\sin(3x^6)}{x^6}}{\lim_{x \rightarrow 0} \frac{\tan(x^3)}{x^3} \cdot \lim_{x \rightarrow 0} \frac{\tan(x^3)}{x^3}} \\ &= \frac{3}{1 \cdot 1} = \frac{3}{1} = 3 \end{aligned}$$

$$\begin{aligned}
 Q_5) \lim_{x \rightarrow 3} \frac{\sqrt{x+1} - 2}{x-3} \cdot \frac{\sqrt{x+1} + 2}{\sqrt{x+1} + 2} \\
 = \lim_{x \rightarrow 3} \frac{x+1-4}{(x-3)(\sqrt{x+1} + 2)} = \lim_{x \rightarrow 3} \frac{\cancel{x-3}}{\cancel{(x-3)}(\sqrt{x+1} + 2)} \\
 = \frac{1}{2+2} = \frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 Q_6) \tan^{-1}\left(\lim_{x \rightarrow \infty} \frac{1}{x^2+1}\right) &= \tan^{-1}\left(\frac{1}{\infty}\right) = \tan^{-1}(0) \\
 &= 0 + n\pi; n=0, \pm 1, \pm 2, \dots
 \end{aligned}$$

$$\begin{aligned}
 Q_7) x^2 + e^x > 0 \text{ always} \\
 \text{Therefore } \ln(x^2 + e^x) \text{ cts everywhere.}
 \end{aligned}$$

$$\begin{aligned}
 Q_8) f(0) &= 0 \\
 \lim_{x \rightarrow 0} f(x) &\begin{cases} \rightarrow \lim_{x \rightarrow 0^+} e^{2x} - 1 = e^0 - 1 = 1 - 1 = 0 \\ \rightarrow \lim_{x \rightarrow 0^-} 0 = 0 \end{cases}
 \end{aligned}$$

$$\therefore \lim_{x \rightarrow 0} f(x) = 0$$

$$\Rightarrow f(0) = \lim_{x \rightarrow 0} f(x)$$

f is cts at $x=0$

1-11-2016

Q1) Circle the correct answer:

1- Given that $f(x) = \frac{x^2+6}{x^2-2x}$ and $g(x) = \frac{1}{x^2-1}$, then $D_{g \circ f}$ is

2- Let $g(x)$ be the function obtained from $f(x) = |x| + x^2$ by shifting 3 units to the right, 2 units up and reflected about the x-axis, then $g(4) =$

a) -8 b) -2 c) -14 d) -4

3- If $f(x) = \frac{\cos x}{|x|-4}$, then the domain of $f(x)$ is

a) $\mathbb{R} - 4$ b) $\mathbb{R} - (4, 5]$ c) $\mathbb{R} - [4, 5)$ d) $\mathbb{R} - [4, 5]$

4- $f(x) = \ln(1 + \ln x)$, then the domain of f is

5- $f(x) = \ln(1 + \ln x)$, then $f^{-1}(\ln 2) =$

6- If $f(x) = \frac{4}{4+2\sin^3 x}$, then the range of f is

7- $\cot^2(\tan^{-1} x)$

8- If $f(x) = 2x + 1$ and $f(g(x)) = x^3 + x^2 + x + 1$, then $g(-1) =$

a) 14 b) 7 c) $\frac{-1}{2}$ d) $\frac{1}{2}$

Q2) Fill The Following Blank:

If $f(x) = \sqrt{(\ln x)^2 - 3\ln x + 2}$, then the domain of f is

Good Luck

Ahmad Arafah

0786543665

$$Q_1) \quad \square g \circ f = \frac{1}{\left(\frac{x^2+6}{x^2-2x}\right)^{-1}}$$

$$\begin{aligned} D g(f(x)) &= (\mathbb{R} \cap (\mathbb{R} - \{0, 2\})) - \{-3\} \\ &= (\mathbb{R} - \{0, 2\}) - \{-3\} \\ &= \mathbb{R} - \{0, 2, -3\} \end{aligned}$$

أضرب المقام

$$\frac{x^2+6}{x^2-2x} - 1 = 0$$

$$\frac{x^2+6}{x^2-2x} = 1$$

$$x^2+6 = x^2-2x$$

$$-2x = 6$$

$$\Rightarrow x = -3$$

2) $g(x) \Rightarrow$ 3 unit right تأثير داخلي
2 unit up تأثير خارجي
reflection x-axis $-f(x)$

$$g(x) = -(|x-3| + (x-3)^2 + 2)$$

$$g(4) = -(1 + 1 + 2) = -4$$

$$13) \quad [x] - 4 = 0$$

$$[x] = 4, \quad 4 \leq x < 5$$

$$Df(x) = \mathbb{R} - [4, 5)$$

$$4) \quad 1 + \ln x > 0$$

$$\ln x > -1$$

$$e^{\ln x} > e^{-1}$$

$$x > e^{-1}$$

$$Df(x) = (e^{-1}, \infty)$$

$$5) \quad \ln(1 + \ln x) = \ln 2$$

$$e^{\ln(1 + \ln x)} = e^{\ln 2}$$

$$1 + \ln x = 2 \quad \ln x = 1 \Rightarrow e^{\ln x} = e^1$$

$$x = e$$

$$6) -1 \leq \sin^3 x \leq 1$$

$$-2 \leq 2 \sin^3 x \leq 2$$

$$2 \leq 4 + 2 \sin^3 x \leq 6$$

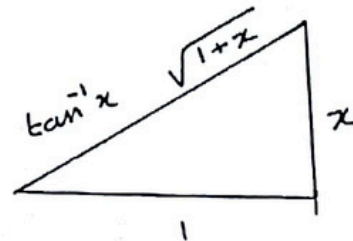
$$\frac{1}{6} \leq \frac{1}{4 + 2 \sin^3 x} \leq \frac{1}{2}$$

$$\frac{4}{6} \leq \frac{4}{4 + 2 \sin^3 x} \leq \frac{4}{2}$$

$$\text{Range} = \left[\frac{2}{3}, 2 \right]$$

$$7) (\cot(\tan^{-1} x))^2$$

$$= \left(\frac{1}{x} \right)^2 = \frac{1}{x^2}$$



$$8) f(g(-1)) = -1 + 1 - 1 + 1 = 0$$

$$f(g(-1)) = 0 \Rightarrow f(x) = 2x + 1$$

$$f(g(-1)) = 2g(-1) + 1$$

$$0 = 2g(-1) + 1 \Rightarrow 2g(-1) = -1$$

$$g(-1) = -\frac{1}{2}$$

OR

$$f(x) = 2x + 1$$

$$f^{-1}(x) = \frac{x-1}{2}$$

$$f(g(x)) = x^3 + x^2 + x + 1$$

$$f^{-1}(f(g(x))) = f^{-1}(x^3 + x^2 + x + 1)$$

$$g(x) = \frac{x^3 + x^2 + x}{2} \Rightarrow g(-1) = \frac{-1 + 1 - 1}{2} = -\frac{1}{2}$$

$$y = 2x + 1$$

$$x = \frac{y-1}{2}$$

$$\frac{x-1}{2} = y$$

$$f^{-1}(x) = \frac{x-1}{2}$$

Q1) $\square g \circ f = \frac{1}{\left(\frac{x^2+6}{x^2-2x}\right) - 1}$

$$\begin{aligned} D g(f(x)) &= (\mathbb{R} \cap (\mathbb{R} - \{0, 2\})) - \{-3\} \\ &= (\mathbb{R} - \{0, 2\}) - \{-3\} \\ &= \mathbb{R} - \{0, 2, -3\} \end{aligned}$$

أضرب المقام

$$\frac{x^2+6}{x^2-2x} - 1 = 0$$

$$\frac{x^2+6}{x^2-2x} = 1$$

$$x^2+6 = x^2-2x$$

$$-2x = 6$$

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2) $g(x) \Rightarrow$ 3 unit right تأثير داخلي
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reflection x-axis $-f(x)$

$$g(x) = -(|x-3| + (x-3)^2 + 2)$$

$$g(4) = -(1+1+2) = -4$$

$$3) [x] - 4 = 0$$

$$[x] = 4, \quad 4 \leq x < 5$$

$$Df(x) = \mathbb{R} - [4, 5]$$

$$4) 1 + \ln x > 0$$

$$\ln x > -1$$

$$e^{\ln x} > e^{-1}$$

$$x > e^{-1}$$

$$Df(x) = (e^{-1}, \infty)$$

$$5) \ln(1 + \ln x) = \ln 2$$

$$e^{\ln(1 + \ln x)} = e^{\ln 2}$$

$$1 + \ln x = 2 \Rightarrow \ln x = 1 \Rightarrow e^{\ln x} = e^1$$

$$x = e$$

Calculus I

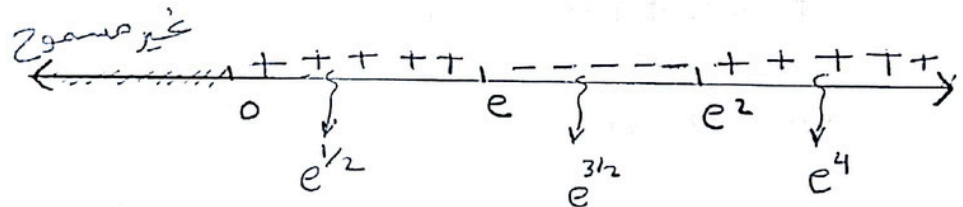
Past Paper

Ahmad Arafah

$$Q_2) \sqrt{(\ln x)^2 - 3 \ln x + 2} = \sqrt{(\ln x - 2)(\ln x - 1)}$$

$$\ln x = 2 \longrightarrow x = e^2$$

$$\ln x = 1 \longrightarrow x = e$$



$$D f(x) = (0, e] \cup [e^2, \infty)$$

Q1) Find the domain of $f(x) = \ln(x+1) + \sqrt{4-x^2}$

Q2) The domain of $f(x) = \sin^{-1}(\ln x) \rightsquigarrow -1 \leq \ln x \leq 1$

$$e^{-1} \leq \ln x \leq e^1$$

$$e^{-1} \leq x \leq e$$

$$[e^{-1}, e]$$

$$\begin{aligned} \text{Q3) } \lim_{x \rightarrow 0} \frac{\sin^2(3x)}{3x^2} &= \lim_{x \rightarrow 0} \frac{\sin(3x)}{3x} \cdot \lim_{x \rightarrow 0} \frac{\sin(3x)}{x} \\ &= 1 \cdot 3 = 3 \end{aligned}$$

$$\text{Q4) } \lim_{x \rightarrow \infty} \sqrt{4x^2 + x} - 2x$$

$$\text{Q5) The limit } \lim_{x \rightarrow 0^-} \frac{x^3 - 1}{x^4 - x^3} = \frac{-1}{0^+ + 0^+} = \frac{-1}{0^+} = -\infty$$

Q6) The number of discontinuities of the function

$$f(x) = \frac{13 - x}{5x^2 + 3 \sin^6 x}$$

equals to

Q7) "Final 2017" $f(x) = \frac{x+1}{x-1}$, $x > 1$, then $f^{-1}(x) =$

$$\text{Q8) "Final 2017" } \lim_{x \rightarrow 2} \frac{\sqrt{x^2 + 12} - 4}{x - 2}$$

$$\text{Q9) "Final 2017" } \lim_{x \rightarrow -1} \frac{\tan(x+1)}{x^2 - 1} = \frac{\lim_{x \rightarrow -1} \frac{\tan(x+1)}{x+1}}{\lim_{x \rightarrow -1} \frac{x^2 - 1}{x+1}}$$

Q10) "Final 2017" The domain of function $f(x) = \cos\left(\frac{1}{\sqrt{4-x^2}}\right)$ is

Q11) "Final 2017" The range of the function $f(x) = 2 - \sqrt{x-2}$

Q12) "Final 2017" If $\ln(\ln(x-1)) = \ln 2 - \ln 3$, then $x =$

Q13) The horizontal asymptote(s) of $f(x) = \frac{\sqrt{x^2+1}}{x-1}$ are

Q14) Mid 2017 $\lim_{x \rightarrow 0} \frac{\sin^9(4x)}{x^9} = \lim_{x \rightarrow 0} \frac{\sin(4x)}{x} \cdot \frac{\sin(4x)}{x} \cdot \frac{\sin(4x)}{x} \cdots \frac{\sin(4x)}{x} \cdot \sin(4x)$
 $= 1 \cdot 1 \cdot 1 \cdots 1 \cdot 0$
 $= 0$

أمثلة سنوادة على النهايات
(بوليتكنك)

Q1) The limit

$$\lim_{x \rightarrow \infty} [\sqrt{x^2 + (a+2)x} - \sqrt{x^2 + 2x}]$$

$$= \frac{a^2}{2}$$

then the value of the constant a equals to

Sol:

$$\lim_{x \rightarrow \infty} [\sqrt{x^2 + (a+2)x} - \sqrt{x^2 + 2x}]$$

$$\frac{\sqrt{x^2 + (a+2)x} + \sqrt{x^2 + 2x}}{\sqrt{x^2 + (a+2)x} + \sqrt{x^2 + 2x}}$$

$$= \lim_{x \rightarrow \infty} \frac{x^2 + ax + 2x - x^2 - 2x}{\sqrt{x^2 + (a+2)x} + \sqrt{x^2 + 2x}}$$

$$= \lim_{x \rightarrow \infty} \frac{ax}{\sqrt{x^2 + (a+2)x} + \sqrt{x^2 + 2x}}$$

$$= \lim_{x \rightarrow \infty} \frac{ax}{|x| + |2|} = \lim_{x \rightarrow \infty} \frac{ax}{2x}$$

$$= \frac{a}{2}$$

$$\Rightarrow \frac{a}{2} = \frac{a^2}{2}$$

$$\Rightarrow a^2 - a = 0 \Rightarrow a(a-1) = 0 \Rightarrow a=1$$

Q2) The H.A for

$f(x) = e^{2-x^2}$ is (are)

$$\text{Sol: } \lim_{x \rightarrow \infty} e^{2-x^2} = e^{-\infty} = 0$$

$$\lim_{x \rightarrow -\infty} e^{2-x^2} = e^{2-(-\infty)^2} = e^{-\infty} = 0$$

$\therefore y=0$ is H.A

$$Q3) \lim_{x \rightarrow \infty} \frac{2x+1}{\sqrt{3x^2+5}-7}$$

$$\text{Sol: } \lim_{x \rightarrow \infty} \frac{2x+1}{\sqrt{3x^2+5}-7} = \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{2x}{\sqrt{3} |x|} = \lim_{x \rightarrow \infty} \frac{2x}{\sqrt{3} x} = \frac{2}{\sqrt{3}}$$

$$Q4) \lim_{x \rightarrow 0} \frac{\sin(2x)}{x + \tan(3x)}$$

$$\text{Sol: } \lim_{x \rightarrow 0} \frac{\sin(2x)}{x + \tan(3x)} = \frac{0}{0}$$

x is small, $\sin(2x) \approx 2x$, $\tan(3x) \approx 3x$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\frac{\sin(2x)}{x}}{\frac{x}{x} + \frac{\tan(3x)}{x}}$$

$$= \frac{2}{1+3} = \frac{2}{4} = \frac{1}{2}$$

$$Q5) \lim_{x \rightarrow \infty} \frac{3x - 5x^3 + 1}{x - x^5 + 1}$$

$$\lim_{x \rightarrow \infty} \frac{-5x^3}{-x^5} = \lim_{x \rightarrow \infty} \frac{5}{x^2}$$

$$= \frac{5}{\infty} = 0$$

Q6) The limit

$$\lim_{x \rightarrow 0} x^2 \left[\sin^2 \left(\cos \left(\frac{x}{2} \right) \right) + \frac{2}{\sin^2(2x)} \right]$$

Sol: →

$$\lim_{x \rightarrow 0} \underbrace{x^2 \sin^2\left(\cos\left(\frac{5}{x}\right)\right)}_{\text{I}} + \underbrace{\frac{2x^2}{\sin^2(2x)}}_{\text{II}}$$

$$\text{I} \quad \lim_{x \rightarrow 0} x^2 \sin^2\left(\cos\left(\frac{5}{x}\right)\right)$$

$$0 \leq \sin^2\left(\cos\left(\frac{5}{x}\right)\right) \leq 1$$

$$0 \leq x^2 \sin^2\left(\cos\left(\frac{5}{x}\right)\right) \leq x^2$$

$$\begin{array}{ccc} & \downarrow & \\ 0 & & 0 \end{array}$$

$$\text{II} \quad \lim_{x \rightarrow 0} \frac{2x^2}{\sin^2(2x)}$$

$$= 2 \left(\lim_{x \rightarrow 0} \frac{x}{\sin(2x)} \right)^2$$

$$= 2 \cdot \left(\frac{1}{2}\right)^2 = \frac{2}{4} = \frac{1}{2}$$

$$\therefore \lim_{x \rightarrow 0} x^2 \sin^2\left(\cos\left(\frac{5}{x}\right)\right) + \frac{2x^2}{\sin^2(2x)}$$

$$= 0 + \frac{1}{2} = \frac{1}{2}$$

Q7) The V.A of the function $f(x) = \frac{2}{x} + \frac{1}{x^2}$ is

Sol: → $x=0$

$$\lim_{x \rightarrow 0^+} \frac{2}{x} + \frac{1}{x^2} = \frac{2}{0^+} + \frac{1}{0^+}$$

$$= \infty + \infty = \infty$$

$\therefore x=0$ is V.A

Q8) The number of H.A for the function $f(x) = \sqrt{x^2 - x} - x^7$ is ?

Sol: →

$$\lim_{x \rightarrow \infty} \sqrt{x^2 - x} - x^7 = \sqrt{\infty + \infty} - (-\infty)^7 = \infty + \infty = \infty \quad \left(\begin{array}{c} \text{No} \\ \text{H.A} \end{array} \right)$$

$$\lim_{x \rightarrow \infty} \sqrt{x^2 - x} - x^7 = \infty - \infty!$$

$$\lim_{x \rightarrow \infty} \sqrt{x^2 - x} - x^7 \cdot \frac{\sqrt{x^2 - x} + x^7}{\sqrt{x^2 - x} + x^7}$$

$$= \lim_{x \rightarrow \infty} \frac{x^2 - x - x^{14}}{\sqrt{x^2 - x} + x^7}$$

$$= \lim_{x \rightarrow \infty} \frac{-x^{14}}{x^7} = \lim_{x \rightarrow \infty} -x^7$$

$$= -\infty \quad \left(\begin{array}{c} \text{No} \\ \text{H.A} \end{array} \right)$$

The number of H.A(s) is

0

Q9) $\lim_{x \rightarrow 1} \frac{e^{2 \ln x} - 9}{x^3 - 3x^2 + x - 3}$

Sol: → $\frac{e^{2 \ln 1} - 9}{1 - 3 + 1 - 3}$

$$= \frac{e^0 - 9}{-4} = \frac{1 - 9}{-4} = \frac{-8}{-4} = 2$$

Q10) $\lim_{x \rightarrow 2^-} \left(\frac{x+2}{2-x} \right) = \frac{4}{0^+} = \infty$

Q.11) $\lim_{x \rightarrow \infty} \sqrt{x^2-5} - x = \infty - \infty!$

Sol: $\lim_{x \rightarrow \infty} \sqrt{x^2-5} - x \cdot \frac{\sqrt{x^2-5}+x}{\sqrt{x^2-5}+x}$

$$= \lim_{x \rightarrow \infty} \frac{x^2-5-x^2}{\sqrt{x^2-5}+x}$$

$$= \lim_{x \rightarrow \infty} \frac{-5}{\sqrt{x^2-5}+x} =$$

$$= \frac{-5}{\infty} = 0$$

Q.12) $\lim_{x \rightarrow \infty} \sqrt{x^2+4x} - \sqrt{x^2+x}$

Sol: $= \infty - \infty!$

$$\lim_{x \rightarrow \infty} \sqrt{x^2+4x} - \sqrt{x^2+x} \cdot \frac{\sqrt{x^2+4x} + \sqrt{x^2+x}}{\sqrt{x^2+4x} + \sqrt{x^2+x}}$$

$$= \lim_{x \rightarrow \infty} \frac{x^2+4x-x^2-x}{\sqrt{x^2+4x} + \sqrt{x^2+x}}$$

$$= \lim_{x \rightarrow \infty} \frac{3x}{\sqrt{x^2+4x} + \sqrt{x^2+x}}$$

$$= \lim_{x \rightarrow \infty} \frac{3x}{|x|+|x|} = \lim_{x \rightarrow \infty} \frac{3x}{2x}$$

$$= \frac{3}{2}$$

Q.13) If $f(x) = \frac{x^2-1}{x^2-2x+1}$, then

H.A and V.A are: $\Rightarrow$

Sol: $\text{H.A } \lim_{x \rightarrow \pm \infty} \frac{x^2-1}{x^2-2x+1}$

$$= \lim_{x \rightarrow \pm \infty} \frac{x^2}{x^2} = 1$$

$\therefore y=1$ is H.A

V.A $x=1$ $\therefore$ $\lim_{x \rightarrow 1} \frac{x^2-1}{x^2-2x+1} = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)(x-1)}$

$$= \lim_{x \rightarrow 1} \frac{x+1}{x-1} = \frac{2}{0} = \pm \infty$$

$\therefore x=1$ is V.A

Q.14) $\lim_{x \rightarrow \infty} \sqrt{x+3\sqrt{x}} - \sqrt{x} = \infty - \infty!$

Sol: $= \infty - \infty!$

$$\Rightarrow \lim_{x \rightarrow \infty} \sqrt{x+3\sqrt{x}} - \sqrt{x} \cdot \frac{\sqrt{x+3\sqrt{x}} + \sqrt{x}}{\sqrt{x+3\sqrt{x}} + \sqrt{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{x+3\sqrt{x}-x}{\sqrt{x+3\sqrt{x}} + \sqrt{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{3\sqrt{x}}{\sqrt{x}(\sqrt{1+\frac{3}{\sqrt{x}}} + 1)}$$

$$= \frac{3}{\sqrt{1+0} + 1} = \frac{3}{2}$$

Q.15) If $\lim_{x \rightarrow 0} \frac{\sqrt{ax+b}-2}{x} = 4$

then the values of a and b are 8

Sol: إذا كانت النهاية موجودة،
في نهاية المقام = صفر فإن
في نهاية البسط = صفر

$$\lim_{x \rightarrow 0} x = 0 \Rightarrow \lim_{x \rightarrow 0} \sqrt{ax+b}-2=0$$

$$\sqrt{b}=2 \Rightarrow b=4$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{ax+4} - 2}{x} \cdot \frac{\sqrt{ax+4} + 2}{\sqrt{ax+4} + 2} \Rightarrow 2 \lim_{x \rightarrow a} g(x) = 1$$

$$= \lim_{x \rightarrow 0} \frac{ax+4-4}{x(\sqrt{ax+4}+2)}$$

$$= \lim_{x \rightarrow 0} \frac{a}{\sqrt{ax+4}+2} = \frac{a}{4}$$

$$\Rightarrow \frac{a}{4} = 1$$

$$\Rightarrow a = 4$$

Q16) IF $\lim_{x \rightarrow a} [f(x) + g(x)] = 3$
and $\lim_{x \rightarrow a} [f(x) - g(x)] = 2$

then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} =$

Sol:

(I) $\lim_{x \rightarrow a} [f(x) + g(x)] = 3$

$\lim_{x \rightarrow a} [f(x) - g(x)] = 2$

$\Rightarrow \lim_{x \rightarrow a} [f(x) + g(x) + f(x) - g(x)] = 5$

$\Rightarrow 2 \lim_{x \rightarrow a} f(x) = 5$

$\Rightarrow \lim_{x \rightarrow a} f(x) = \frac{5}{2}$

(II) $\lim_{x \rightarrow a} [f(x) + g(x)] = 3$

$= 3 - 2$

$\Rightarrow \lim_{x \rightarrow a} [f(x) + g(x) - f(x) + g(x)] = 1$

$$\lim_{x \rightarrow a} g(x) = \frac{1}{2}$$

$\therefore \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{5/2}{1/2} = 5$

Q17) $\lim_{x \rightarrow \infty} \frac{3x^2 + \cos^2 x}{x^2 + 1}$

Sol: $\lim_{x \rightarrow \infty} \frac{3x^2}{x^2} + \frac{\cos^2 x}{x^2}$

$$= \frac{x^2}{x^2} + \frac{1}{x^2}$$

$$= \frac{3+0}{1+0} = 3$$

$+ \lim_{x \rightarrow \infty} \frac{\cos^2 x}{x^2}$

$$0 \leq \cos^2 x \leq 1$$

$$0 \leq \frac{\cos^2 x}{x^2} \leq \frac{1}{x^2}$$

$\downarrow \quad \quad \quad \downarrow$
 $0 \quad \quad \quad 0$

Q18) The H.A(s) of $f(x) = \sqrt{x^2+x+1} - \sqrt{x^2-4x}$ are (How)

Sol:

$y = -\frac{5}{2}, \frac{5}{2}$ are H.A(s)

Q19) Let $f(x) = \frac{5}{x^2 - Kx + \frac{9}{4}}$

then the values of K such that f has no vertical ?

Sol: no vertical = لا يوجد للمقام

لا يكون للمعادلة التربيعية
جذور إذا كان مميزها
أقل من صفر

$$b^2 - 4ac < 0$$

$$K^2 - 4(1)(\frac{9}{4}) < 0$$

$$K^2 - 9 < 0 \Rightarrow K^2 < 9$$

$$\Rightarrow \sqrt{K^2} < \sqrt{9}$$

$$\Rightarrow |K| < 3$$

$$-3 < K < 3$$

$$K \in (-3, 3)$$

Q20) $\lim_{x \rightarrow \infty} \frac{x + \sin 5x}{5x + \sin x}$

Sol: x على السمت

$$\lim_{x \rightarrow \infty} \frac{\frac{x}{x} + \frac{\sin 5x}{x}}{\frac{5x}{x} + \frac{\sin x}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{1 + \frac{\sin 5x}{x}}{5 + \frac{\sin x}{x}} = \frac{1+0}{5+0}$$

$$= \frac{1}{5}$$

$$\begin{array}{l} -1 \leq \sin x \leq 1 \\ \frac{-1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x} \\ \downarrow \quad \downarrow \quad \downarrow \\ 0 \quad 0 \quad 0 \end{array} \quad \begin{array}{l} -1 \leq \sin 5x \leq 1 \\ \frac{-1}{x} \leq \frac{\sin 5x}{x} \leq \frac{1}{x} \\ \downarrow \quad \downarrow \quad \downarrow \\ 0 \quad 0 \quad 0 \end{array}$$

Q21) The vertical asy. of $f(x) = \ln(x-2)$ is

Sol: $x=2$

$$\lim_{x \rightarrow 2^+} \ln(x-2) = \ln 0^+ = -\infty$$

$$x=2 \text{ V.A}$$

Q22) $\lim_{x \rightarrow \frac{3\pi}{2}^+} \sin x$

$$\begin{array}{c} \pi \quad \frac{3\pi}{2} \quad 2\pi \\ \text{Sol: } \lim_{x \rightarrow \frac{3\pi}{2}^+} -\sin x = -1 \end{array}$$

Q23) $\lim_{x \rightarrow 2^+} (x-2) \sin\left(\frac{1}{x^2-4}\right)$

Sol: $= 0$ (dnc)!

$$-1 \leq \sin\left(\frac{1}{x^2-4}\right) \leq 1$$

$$-(x-2) \leq (x-2) \sin\left(\frac{1}{x^2-4}\right) \leq (x-2)$$

$$\lim_{x \rightarrow 2^+} -(x-2) = 0$$

$$\lim_{x \rightarrow 2^+} (x-2) = 0$$

0

Q24) If $f(x) = \frac{2x^2+1}{(2x-1)^2-1}$, then

H.A and V.A are $\rightarrow$
(H.W)

Sol: $\rightarrow$ H.A $y = \frac{1}{2}$
V.A $x=1$ and $x=0$

Q25) $\lim_{x \rightarrow 3} (x^2-9) \cot(x^2-x-6)$

Sol: $\rightarrow 0 \cdot \infty$!

$$\lim_{x \rightarrow 3} (x^2-9) \frac{\cos(x^2-x-6)}{\sin(x^2-x-6)}$$

$$= \lim_{x \rightarrow 3} \cos(x^2-x-6) \cdot \lim_{x \rightarrow 3} \frac{x^2-9}{\sin(x^2-x-6)}$$

$$= 1 \cdot \lim_{x \rightarrow 3} \frac{x^2-9}{\frac{\sin(x^2-x-6)}{x^2-x-6}}$$

$$= \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{(x-3)(x+2)} = \frac{1}{1}$$

$$= \frac{5}{6}$$

Q26) $\lim_{x \rightarrow -\infty} \sqrt{x^2+4x} - \sqrt{x^2-x}$

Sol: $\rightarrow \lim_{x \rightarrow -\infty} \sqrt{x^2+4x} - \sqrt{x^2-x}$

$$\frac{\sqrt{x^2+4x} + \sqrt{x^2-x}}{\sqrt{x^2+4x} + \sqrt{x^2-x}}$$

$$= \lim_{x \rightarrow -\infty} \frac{5x}{\sqrt{x^2+4x} + \sqrt{x^2-x}}$$

$$= \lim_{x \rightarrow -\infty} \frac{5x}{|x| + |x|}$$

$$x \rightarrow -\infty, |x| \rightarrow -x$$

$$= \lim_{x \rightarrow -\infty} \frac{5x}{-2x} = -\frac{5}{2}$$

Q27) If $\lim_{x \rightarrow 1} \frac{x^3+2x^2-x+a}{x^2-1}$ exists,

then $a = ?$

Sol: $\rightarrow$ بما أن النهاية موجودة ونسبة
المقام = صفر $\leftarrow$ نسبة البسط = صفر

$$\Rightarrow \lim_{x \rightarrow 1} \frac{x^3+2x^2-x+a}{x^2-1} = 0$$

$$1+2-1+a=0$$

$$\therefore a = -2$$

Q28) a) $\lim_{x \rightarrow 0} \sqrt{x+9} - 3$ (H.W)

Sol: $\rightarrow \frac{1}{6}$

b) $\lim_{x \rightarrow 0^-} 3x \sin\left(\frac{2}{x}\right)$ [H.W]

Sol: $\rightarrow 0$

Q21) H.A for

$$f(x) = \frac{2x^2+1}{x+2x^2} + \frac{x-2}{3-|x|}$$

$$\text{Sol: } \lim_{x \rightarrow \infty} \frac{2x^2+1}{x+2x^2} + \frac{x-2}{3-|x|}$$

$$= \lim_{x \rightarrow \infty} \frac{2x^2}{2x^2} + \frac{x}{-x}$$

$$= 1 - 1 = 0$$

$$\therefore y = 0 \text{ H.A.}$$

$$\text{② } \lim_{x \rightarrow \infty} \frac{2x^2}{2x^2} + \frac{x}{x}$$

$$= 1 + 1 = 2$$

$$\therefore y = 2 \text{ H.A.}$$

$y = 0, 2$ are H.A.

$$Q30) \lim_{x \rightarrow 3^-} \frac{x^2-3x}{x^2-6x+9}$$

$$\text{Sol: } = \frac{0}{0}!$$

$$\lim_{x \rightarrow 3^-} \frac{x(x-3)}{(x-3)(x-3)} = \frac{3}{0^-} = -\infty$$

$$Q31) \lim_{x \rightarrow 3} \frac{\sin(x-3)}{x^2-2x-3}$$

$$\text{Sol: } = \frac{0}{0}!$$

$$\lim_{x \rightarrow 3} \frac{\sin(x-3)}{(x-3)(x+1)}$$

$$= \lim_{x \rightarrow 3} \frac{\sin(x-3)}{x-3} \cdot \lim_{x \rightarrow 3} \frac{1}{x+1}$$

$$= 1 \cdot \frac{1}{4} = \frac{1}{4}$$

$$Q32) \lim_{x \rightarrow \infty} \sqrt{ax^2+bx} - 3x = 2$$

6/12/2016

then the values of a and b are :-

$$\text{Sol: } \lim_{x \rightarrow \infty} \sqrt{ax^2+bx} - 3x = 2$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{ax^2+bx-9x^2}{\sqrt{ax^2+bx}+3x} = 2$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{(a-9)x^2+bx}{\sqrt{ax^2+bx}+3x} = 2$$

بما أن النهاية موجودة وتساوي عدد ثابت
يجب أن تكون أعلى قوة في البسط = أعلى
تو في المقام. لهذا يجب أن يكون
مسا على x^2 في البسط = صفر

$$\Rightarrow a = 9$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{bx}{3x+3x} = 2$$

$$\lim_{x \rightarrow \infty} \frac{bx}{6x} = 2$$

$$\frac{b}{6} = 2 \Rightarrow b = 12$$

Q33) $\lim_{x \rightarrow 1} \frac{(x-1)}{\sqrt{x^2-2x+1}}$

2016

Sol: $\frac{0}{0}!$

$$\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{(x-1)^2}} = \lim_{x \rightarrow 1} \frac{(x-1)}{|x-1|}$$

--- + + + + |x-1|

⊕ $\lim_{x \rightarrow 1^+} \frac{x-1}{x-1} = 1$

⊖ $\lim_{x \rightarrow 1^-} \frac{x-1}{-(x-1)} = -1$ $\Rightarrow$

$\lim_{x \rightarrow 1} \frac{x-1}{|x-1|}$ d.n.e.

Q34) $\lim_{x \rightarrow \infty} 3x \tan\left(\frac{2}{x}\right)$

2015

Sol: $\infty \cdot 0!$ أسهل

$y = \frac{1}{x} \Rightarrow x = \frac{1}{y}$

$x \rightarrow \infty, y \rightarrow 0$

$\lim_{y \rightarrow 0} \frac{3 \tan(2y)}{y}$

$= 3 \lim_{y \rightarrow 0} \frac{\tan(2y)}{y}$

$= 3 \cdot \frac{2}{1} = 6$

Q35) The H.A and V.A for

2016

$f(x) = e^{\frac{(x+\sin 3x)}{2x}}$

Sol: V.A $x=0$

⊕ $\lim_{x \rightarrow 0} e^{\frac{(x+\sin 3x)}{2x}}$

$= e^{\lim_{x \rightarrow 0} \left(\frac{x}{2x} + \frac{\sin 3x}{2x} \right)}$

$= e^{\frac{1}{2} + \frac{3}{2}} = e^2 \neq \pm \infty$

$\therefore x=0$ (Not V.A)

⊕ $\lim_{x \rightarrow \pm \infty} e^{\frac{(x+\sin 3x)}{2x}}$

$= e^{\lim_{x \rightarrow \pm \infty} \left(\frac{x}{2x} + \frac{\sin 3x}{2x} \right)}$

$= e^{\frac{1}{2} + 0}$

$= \sqrt{e}$

$\therefore y = \sqrt{e}$ H.A

* حل أسئلة سينوس

$$Q) \lim_{x \rightarrow 0} \cos \left(\frac{\sin(x+\pi)}{2x} \right)$$

$$\underline{\underline{\text{Sol.}}} \cos \left(\lim_{x \rightarrow 0} \frac{\sin(x+\pi)}{2x} \right)$$

$$\cos \left(\lim_{x \rightarrow 0} \frac{\sin x \cos \pi + \sin \pi \cos x}{2x} \right)$$

$$\cos \left(\lim_{x \rightarrow 0} \frac{-\sin x}{2x} \right) = \cos \left(\frac{-1}{2} \right)$$

$$= \cos \left(\frac{1}{2} \right)$$

Q) The limit

$$\lim_{x \rightarrow 2} 3 \sin^{-1} \left(\frac{x^2 - 5x + 6}{x^2 - 2x} \right)$$

Sol.

$$3 \sin^{-1} \left(\lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x^2 - 2x} \right)$$

$$3 \sin^{-1} \left(\lim_{x \rightarrow 2} \frac{(x-3)(x-2)}{x(x-2)} \right)$$

$$= 3 \sin^{-1} \left(\frac{-1}{2} \right) = -3 \sin^{-1} \left(\frac{1}{2} \right)$$

$$= -3 \cdot \frac{\pi}{6} = -\frac{\pi}{2}$$

H.W) Q) $f(x) = \begin{cases} \frac{\tan kx}{x}, & x < 0 \\ 2x + 3k^2, & x \geq 0 \end{cases}$

all values of k that let $\lim_{x \rightarrow 0} f(x)$ exists are

Sol. $k = 0, \frac{1}{3}$

2) $\lim_{x \rightarrow 0} \tan \left(\frac{1 - \cos x}{x} \right)$

Sol. $\tan \left(\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} \right)$

$$\tan \left(\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} \cdot \frac{1 + \cos x}{1 + \cos x} \right)$$

$$= \tan \left(\lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{2x} \right)$$

$$= \tan \left(\lim_{x \rightarrow 0} \frac{\sin^2 x}{2x} \right)$$

$$= \tan \left(\lim_{x \rightarrow 0} \frac{\sin x}{2x} \cdot \lim_{x \rightarrow 0} \sin x \right)$$

$$= \tan(0) = 0$$

Q) Let $f(x) = \sqrt{|x| - 4}$, then f continuous on

Sol. $|x| - 4 \geq 0$

$$|x| \geq 4 \Rightarrow x \geq 4 \text{ or } x \leq -4$$

$$[4, \infty), (-\infty, -4]$$

Q) All values of a such that

$$f(x) = \begin{cases} x+4, & x \leq a \\ x^2, & x > a \end{cases}$$

cts on $\mathbb{R}$ are

Sol. $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$

$$\lim_{x \rightarrow a^+} x^2 = \lim_{x \rightarrow a^-} x+4$$

$$a^2 = a+4$$

$$a^2 - a - 4 = 0$$

$$a = \frac{1 \pm \sqrt{1 - 4(1)(-4)}}{2(1)} = \frac{1 \pm \sqrt{17}}{2}$$

Q) $f(x) = \csc x$ is discontinuous at $|x-1| = 2$

Sol.

$$f(x) = \frac{1}{\sin x (|x-1| - 2)}$$

$$\sin x = 0, x = \pm n\pi \quad (\text{مواقع التوقف})$$

$$|x-1| - 2 = 0 \Rightarrow x-1 = 2 \Rightarrow x = 3$$

$$\Rightarrow x-1 = -2 \Rightarrow x = -1$$

discts. at $x = \pm n\pi, -1, 3$

Q₁: Simplify the expression

$$\cos(2\tan^{-1}x)$$

Q₂: Find $(f \circ g)(x)$ if

$$f(x) = x^2 - 1, g(x) = 2x + 1$$

Sol $f(g(x)) = 4x^2 + 4x$

Q₃: The domain of

$$f(x) = \sqrt{4-x^2} + \frac{1}{x-2} \text{ is?}$$

Sol: $[-2, 2)$

Q₄: Find $(f \circ g \circ h)(x)$:

$$f(x) = x + 1, g(x) = 2x,$$

$$h(x) = x - 1$$

Sol $-1 + 2x$

Q₅: The domain of

$$h(x) = \sqrt{4-x^2} \sin^{-1}(2x-4)$$

is :

Sol $[\frac{3}{2}, 2]$

Q₆: IF $f(x) = \frac{\cos^{-1}(1-x)}{2-x}$,

then the domain is :

Sol: $[0, 2)$.

Q₇: let (x) be a number in the interval $[-1, 1]$ then $\csc(\sin^{-1}x)$ is

Sol $\frac{1}{x}$

Q₈: IF $f(x) = \begin{cases} 2x^2 + 4 & , x \leq 1 \\ x - 1 & , 1 < x \leq 4 \\ x + 1 & , x > 4 \end{cases}$

then $f \circ f(1)$ is

Sol 7

Q₉: IF $\frac{3\pi}{2} < \theta < 2\pi$ and

$$\cos \theta = \frac{1}{4}, \text{ then } \csc \theta = ?$$

Sol $-\frac{4}{\sqrt{15}}$

Q₁₀: Find the Range of

$$f(x) = 5\sin^2 x + 2 \text{ is}$$

Sol $[2, 7]$

Q₁₁: The inverse function of

$$f(x) = \frac{1}{3 + 3\tan^{-1}x}$$

$$3 + 3\tan^{-1}x$$

Sol: $f^{-1}(x) = \tan\left(\frac{1-3x}{3x}\right)$

Q₁₂: The range of

$$f(x) = \frac{6x-2}{3x+1} \text{ is}$$

Sol $\mathbb{R} - \{2\}$.

Q₁₃: which of the following
an even function?

a) $x^2 + 3x$ b) $2 - 3x$

c) $1 + 2\cos x$ d) $\frac{x^3 - x}{1 + x^2}$

Q₁₄: The domain of
 $f(x) = 2 \sin^{-1}(1-2x)$ is

Sol $[0, 1]$.

Q₁₅: The domain of

$$f(x) = \frac{1}{\frac{2}{x-1} - 1} \text{ is}$$

Sol $\mathbb{R} - \{1, 3\}$.

Q₁₆: IF $f(x) = x^3 + 2x + 5$
then $f^{-1}(8) =$

Sol $= 1$.

Q₁₇: The range of

$$g(x) = -x^2 + 2x - 5 \text{ is}$$

Sol $(-\infty, -4]$.

Q₁₈: IF $f(x) = 3x - 5$ and
 $(f \circ g)(x) = 12x^2 - 6x + 13$,
then $g(0) = ?$

Sol 6

Q₁₉: IF $f(x) = \frac{5}{x-3}$ and $g(x) = x^2 - 1$,
then the domain of
 $(f \circ g)(x)$ is

Sol $\mathbb{R} - \{-2, 2\}$.

Q₂₀: The Function $g(x)$ is
obtained from $f(x) = x^2 - 2x$
by shifting $f(x)$ one unit
to the left then reflecting
about the y-axis, then
 $g(x)$ is?

Sol $g(x) = x^2 - 1$

Q₂₁) $\sin(\tan^{-1} 2)$

Sol: $\frac{2}{\sqrt{5}}$

Q22) The domain of $f(x) = 4\sin^{-1}(4x+1)$ is

- a) $[-0.5, 0]$ b) $[-1, 1]$
c) $[-1, 0]$ d) $[-2, 2]$

Q23) Let $f(x) = 3 + 7\sqrt{4-x}$, If $g(x)$ is a function such that the domain of g is $(0, \infty)$, then the domain of multiplication of the functions $f \cdot g(x)$ is

- a) $(0, 4)$ b) $(0, 4]$
c) $(0, \infty)$ d) $(4, \infty)$

Q24) Assume $g(x) = x^2 + 5$, Range $f(x) = [2, 5]$
Find Range $g \circ f(x)$?

- a) $[9, 30]$ b) $(9, 30]$
c) $(9, 30)$ d) $[2, 5]$

Q25) Let $f(x) = \frac{3}{x-3}$ and $g(x) = \frac{2}{x}$,
Then $D_{g \circ f(x)}$ is
Sol: $\mathbb{R} - \{3\}$

Q26) The domain of $f(x) = \frac{\cos x}{1 - \sin x}$ is

Sol: $\mathbb{R} - \left\{ \frac{\pi}{2} + 2n\pi, n=0, 1, 2, \dots \right\}$

Q27) The inverse function of
 $f(x) = 8 + \sqrt[3]{x-4}$

Sol: $(x-8)^3 + 4$

Q28) $\cot(\sin^{-1} \frac{3}{7})$

Sol: $\frac{2\sqrt{10}}{3}$

Q29) $\cos(\arcsin(\frac{1}{4}))$ is

Sol: $\frac{\sqrt{15}}{4}$

Hint: $\arcsin x = \sin^{-1} x$

Q30) The range of $g(x)$ is $[-5, 5]$ and $f(x) = g^2(x)$, then the range of $f(x)$ is

Sol: $[0, 25]$

Q31) The range of $f(x) = 2 + \sqrt{x-1}$

- a) $(2, \infty)$ b) $[2, \infty)$
c) $(-\infty, 2)$ d) $(-\infty, -2]$

Q32) The domain of
 $f(x) = \sqrt{2 - |x+3|}$ is

Sol: $[-5, -1]$

Q33) which of the following function is Symmetric about the origin?

- a) $|x| + \sin x$ b) $x \sin x$
c) $x \cos x$ d) $|x|$

Q34) If $f(x) = \frac{1+x}{1-x}$ and $g(x) = \frac{x}{1-x}$

Find $D \log(x)$?

Sol. $\mathbb{R} - \{\frac{1}{2}, 1\}$

Q35) The equation of $\tan y = 1+x^2$
Symmetric about e.

Sol. y-axis only

Q36) If the domain of $f(x)$ is $[-5, 5]$
and the range is $[1, 4]$, then the
domain of $3\sqrt{f(2x-3)} + 1$ is e.

Sol. $[-1, 4]$

Q37) If $f(x) = \sqrt{x}$, $g(x) = x^3 + 1$,
Find $g(f(4))$?

Sol. 9

Q38) The domain of $f(x) = \frac{1}{(x-3)^5}$

Sol. $\mathbb{R} - \{3\}$

Q39) Find the value of $\log_a h(x)$
If $f(x) = \sqrt{x}$, $g(x) = \frac{1}{x}$, $h(x) = x^3$

Sol. $x^{-3/2}$

Q40) If $g(x) = \sqrt{x}$, Find
 $g(\frac{1}{\sqrt{x}})$?

Sol. $x^{-1/4}$

Q41) The value of $\sin(\frac{5\pi}{6})$

Sol. $\frac{1}{2}$

Q42) The value of $\cos(\frac{4\pi}{3})$

Sol. $-\frac{1}{2}$

Q43) If $f(x) = \sin x$, $g(x) = x^2 + 3$
Find $g(f(\pi))$

Sol. 3

Q44) The domain of
 $f(x) = \sqrt{9-x^2} \cdot \sqrt{x-1}$

Sol. $[1, 3]$

Q45) If $f(x) = 5x - 1$
 $g(x) = \frac{x}{2+x}$, Find $f(g(1))$

Sol. $\frac{2}{3}$

Q46) The domain of $f(x) = \frac{\sqrt{x-2} + 4}{x-5}$

Sol. $[2, \infty) - \{5\}$

Q52) Find domain $g \circ g(x)$

if $g(x) = \sqrt[3]{1-x}$

Sol. $\mathbb{R}$

Q47) If $f(x) = x^3 + x + a$, an odd function, find a ?

Sol. $a = 0$

Q53) Find the value of $\sin(\cos^{-1}(\frac{3}{5})) + \tan(\cos^{-1}(\frac{3}{5}))$

Sol. $\frac{4}{5} + \frac{4}{3} = \frac{32}{15}$

Q48) Express the function in the form $f \circ g \circ h(x)$
 $H(x) = \sqrt[8]{2+1x}$

Sol. $f(x) = \sqrt[8]{x}$, $g(x) = 2+x$, $h(x) = 1x$

Q54) Find $g(x)$ if $f(x) = \frac{x}{x+1}$ and $f \circ g(x) = \frac{1}{x}$?

Sol. $g(x) = \frac{1}{x-1}$

Q49) Express the function in the form $f \circ g \circ h(x)$:
 $H(x) = \sec^4 \sqrt{x}$

Sol. $f(x) = x^4$, $g(x) = \sec x$, $h(x) = \sqrt{x}$

Q55) One of the following functions is not one-to-one:

- a) $f(x) = \tan^{-1}x$ b) $f(x) = |x+1| + 2$
c) $f(x) = x^3$

Q50) $\sin(2 \cos^{-1}(\frac{3}{5}))$

Sol. $\frac{24}{25}$

Q56) The graph of $y = x^2 \tan x$ is symmetric about Sol. origin

Q51) The figure shows the graph of (sin) function for one complete period, which of the following is an equation for the graph?

- a) $y = 2 \sin(\frac{x}{2})$ b) $y = 2 \sin x$
c) $y = 2 \sin(2x)$ d) $y = \sin(\pi x)$

Q57) Let $f(x) = \frac{3}{x-1}$ and

$g(x) = 4x$, then the solution(s) of $f(g(x)) = g(f(x))$ is (are)?

- a) 2 b) $\frac{1}{5}$
c) $\frac{1}{5}, 2$ d) 5

Q58) Is the function $f(x) = \frac{\sin(x^2)}{x^3 - 2x}$

odd, even or neither?

Sol. odd.

Q59) Find the inverse of f

$$y = -\sqrt{\frac{2}{x} - 1}$$

Sol. $f^{-1}(x) = \frac{2}{x^2 + 1}$

Q60) Find the domain of f

$$f(x) = \frac{5x+4}{|5x+4|}$$

Sol. $(-\infty, \frac{4}{5}) \cup (\frac{4}{5}, \infty)$

Q61) Find the inverse of f

$$f(x) = (3x-2)^5 + 7$$

Sol. $f^{-1}(x) = \frac{\sqrt[5]{x-7} + 2}{3}$

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